



9709 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the mass-metallicity relation

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	TOI-1420 NIRSpec	NIRSpec Bright Object Time Series	(1) TOI-1420
	2	TOI-1173 b SOSS	NIRISS Single-Object Slitless Spectroscopy	(2) TOI-1173b
	3	TOI-1173 b NIRSpec	NIRSpec Bright Object Time Series	(2) TOI-1173b

Folder	Observation	Label	Observing Template	Science Target
	4	TOI-2134 c SOSS	NIRISS Single-Object Slitless Spectroscopy	(3) TOI-2134_c
	5	TOI-2134 c NIRSpec	NIRSpec Bright Object Time Series	(3) TOI-2134_c
	6	WASP-193 SOSS	NIRISS Single-Object Slitless Spectroscopy	(4) WASP-193
	7	WASP-127b NIRSpec	NIRSpec Bright Object Time Series	(5) WASP-127
	8	KELT-11 SOSS	NIRISS Single-Object Slitless Spectroscopy	(6) KELT-11
	16	KELT-11 NIRSpec	NIRSpec Bright Object Time Series	(6) KELT-11
	9	WASP-131 SOSS	NIRISS Single-Object Slitless Spectroscopy	(7) WASP-131
	10	WASP-131 NIRspec	NIRSpec Bright Object Time Series	(7) WASP-131
	11	WASP-20 SOSS	NIRISS Single-Object Slitless Spectroscopy	(8) WASP-20
	12	WASP-20 NIRSpec	NIRSpec Bright Object Time Series	(8) WASP-20
	13	HD 209458b SOSS	NIRISS Single-Object Slitless Spectroscopy	(9) HD-209458b
KELT-4				
	15	KELT-4 NIRSpec	NIRSpec Bright Object Time Series	(10) KELT-4
	17	NIRISS-Offset_test_15	NIRISS Single-Object Slitless Spectroscopy	(10) KELT-4
	14	KELT-4 SOSS	NIRISS Single-Object Slitless Spectroscopy	(10) KELT-4

ABSTRACT

JWST has revolutionized exoplanet atmospheres, with the transformative capability to measure precise transmission spectra over a broad wavelength range into the IR. These measurements are being used to understand the complex atmospheric physics including the dynamics, cloud, and haze formation across a wide range of physical conditions and to constrain planet formation and evolution using the chemical abundances as primordial tracers. However, clouds have hindered these efforts. Recently, a theoretically predicted large cloud-asymmetry bias has been revealed for a hot Jupiter. Measuring the planet’s limb-asymmetric transmission spectra, the morning limb is seen to be very cloudy while the evening limb is cloud-free – with the re-evaluated metallicity nearly 2 dex lower than a traditional analysis. This huge and prevalent asymmetry bias means most everything we know from transit-derived abundances could be heavily biased. Here we propose the first comprehensive limb-resolved panchromatic survey of gas giant exoplanets. With this survey, the cloudy morning terminator will be fully disentangled from the evening, and accurate unbiased abundances will be measured. Fortuitously, the top limb-asymmetry targets also span two critical parameters—temperature and mass—enabling a broad, efficient statistical survey to: test the hypothesis that diurnal cloud cycling causes widespread limb-asymmetries in transiting exoplanets, and measure the mass-metallicity trend for gas-giants using accurate unbiased metallicities. This program will make a significant impact on both the atmospheric dynamics of exoplanets, and the field of planet formation.

OBSERVING DESCRIPTION

For all targets, we calculated the optimal detector readout patterns and groups per integration using STScI's ExoCTK (<https://exoctk.stsci.edu>) and the official JWST ETC (<https://jwst.etc.stsci.edu/>). The integrations per exposure required to cover each transit event and the signal-to-noise ratios achieved on the exoplanet transmission spectra were calculated using PandEXO (<https://exoctk.stsci.edu/pandexo/>).

NIRSpec G395H: We will use the NIRSpec G395H/F290LP, using the 1.6"×1.6" fixed slit aperture with the Bright Object Time Series (BOTS) mode optimized for transit exoplanets enabling high photometric precision time-series spectroscopy. The SUB2048 subarray will be used to improve duty cycles and capture the full 2.8 to 5.1 μm range at a resolution of $R\sim 3000$. All of our targets will use the NRSRAPID readout pattern, with the groups and integrations optimized using PandEXO.

NIRISS SOSS: We will use the NIRISS GR700XD with the SUBSTRIP256 subarray, which provides broad 0.6-2.8 μm coverage at a $R\sim 700$ with the Bright Object Time Series (BOTS) mode optimized for transit exoplanets enabling high photometric precision time-series spectroscopy. We will make use of the multi-stripe readout patterns for bright targets, to avoid saturation. The groups and readout patterns were optimized with the ETC.

To avoid spectra contamination of nearby stars, we have added position angle constraints to the observations using the ExoCTK tool (https://exoctk.stsci.edu/contam_visibility), which indicates all of our targets will be observable largely free of contamination.

Proposal 9709 - Targets - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the mass-m...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	TOI-1420	RA: 21 31 45.9171 (322.9413212d) Dec: +66 20 55.93 (66.34887d) Equinox: J2000	Proper Motion RA: 45.482 mas/yr Proper Motion Dec: 31.874 mas/yr Parallax: 0.0049134" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]				
	(2)	TOI-1173b	RA: 13 10 43.7621 (197.6823421d) Dec: +70 46 6.27 (70.76841d) Equinox: J2000	Proper Motion RA: 57.787 mas/yr Proper Motion Dec: -92.39299997716444 mas/yr Parallax: 0.0075664" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems, G stars]				
	(3)	TOI-2134_c	RA: 18 07 44.4478 (271.9351992d) Dec: +39 04 26.92 (39.07414d) Equinox: J2000	Proper Motion RA: 54.536 mas/yr Proper Motion Dec: -283.050999905754 mas/yr Parallax: 0.0441087" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. J 6.776 [0.023] C 2003yCat.2246....0C H 6.247 [0.021] C 2003yCat.2246....0C K 6.091 [0.017] C 2003yCat.2246....0C Category=Star Description=[Exoplanet Systems]				
	(4)	WASP-193	RA: 10 57 23.8522 (164.3493842d) Dec: -29 59 49.67 (-29.99713d) Equinox: J2000	Proper Motion RA: -49.055 mas/yr Proper Motion Dec: 1.506 mas/yr Parallax: 0.0026475" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. H 10.810 Category=Star Description=[Exoplanet Systems]					

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(5)	WASP-127	RA: 10 42 14.0837 (160.5586821d) Dec: -03 50 6.26 (-3.83507d) Equinox: J2000	Proper Motion RA: 19.133 mas/yr Proper Motion Dec: 17.058 mas/yr Parallax: 0.0062241" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]			
(6)	KELT-11	RA: 10 46 49.7402 (161.7072508d) Dec: -09 23 56.50 (-9.39903d) Equinox: J2000	Proper Motion RA: -79.16 mas/yr Proper Motion Dec: -78.50699996652111 mas/yr Parallax: 0.010004899999999999" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]			
(7)	WASP-131	RA: 14 00 46.4448 (210.1935200d) Dec: -30 35 0.85 (-30.58357d) Equinox: J2000	Proper Motion RA: 10.24 mas/yr Proper Motion Dec: -9.20499992389523 mas/yr Parallax: 0.0051765" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]			
(8)	WASP-20	RA: 00 20 38.5350 (5.1605625d) Dec: -23 56 8.60 (-23.93572d) Equinox: J2000	Proper Motion RA: 1.352 mas/yr Proper Motion Dec: -14.739000062036212 mas/yr Parallax: 0.0034699" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]			
(9)	HD-209458b	RA: 22 03 10.7727 (330.7948863d) Dec: +18 53 3.55 (18.88432d) Equinox: J2000	Proper Motion RA: 29.765999999999995 mas/yr Proper Motion Dec: -17.976000003727677 mas/yr Parallax: 0.0207694" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]			

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(10)	KELT-4	RA: 10 28 15.0111 (157.0625463d) Dec: +25 34 23.47 (25.57319d) Equinox: J2000	Proper Motion RA: 14.118341354376284 mas/yr Proper Motion Dec: -12.11922514602998 mas/yr Parallax: 0.00458" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Parallax, PM from Gaia DR3 Category=Star Description=[Exoplanet Systems]			
(11)	TOI_2134c_NIRSpec_TA	RA: 18 07 44.5240 (271.9355167d) Dec: +39 03 58.29 (39.06619d) Equinox: J2000	Proper Motion RA: 1.111 mas/yr Proper Motion Dec: 1.919 mas/yr Parallax: 0.000715" Epoch of Position: 2015.5
Comments: Gaia DR3 4610264697788666368 Teff = 6232 K G = 14.108 Category=Star Description=[G stars] Extended=NO			
(12)	WASP-127_NIRSpec_TA	RA: 10 42 11.4176 (160.5475733d) Dec: -03 50 13.04 (-3.83696d) Equinox: J2000	Proper Motion RA: 18.769 mas/yr Proper Motion Dec: 16.489 mas/yr Parallax: 0.006214000000000001" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. J 9.927 Category=Star Description=[K stars]			
(13)	KELT_11_NIRSpec_TA	RA: 10 46 50.2250 (161.7092708d) Dec: -09 23 8.50 (-9.38569d) Equinox: J2000	Proper Motion RA: 5.106 mas/yr Proper Motion Dec: 3.614 mas/yr Parallax: 0.001257" Epoch of Position: 2015.5
Comments: Gaia DR3 3761497933674720000 G = 14.197 J = 13.179 Teff=5838K Category=Star Description=[G stars] Extended=NO			

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(14)	WASP_131_NIRSpec_TA	RA: 14 00 45.5810 (210.1899208d) Dec: -30 35 37.91 (-30.59386d) Equinox: J2000	Proper Motion RA: -20.572 mas/yr Proper Motion Dec: -11.404 mas/yr Parallax: 0.001990" Epoch of Position: 2015.5 Comments: Gaia DR3 6173451442831776896 Teff = 3906 K G = 16.637 J = 14.305 Category=Star Description=[K stars] Extended=NO
(15)	KELT_4_NIRSpec_TA	RA: 10 28 12.5670 (157.0523625d) Dec: +25 34 58.51 (25.58292d) Equinox: J2000	Proper Motion RA: -12.827 mas/yr Proper Motion Dec: -0.697 mas/yr Parallax: 0.000714" Epoch of Position: 2015.5 Comments: Gaia DR3 727624226525958912 Teff = 5920 K G = 14.93 J = 13.953 Category=Star Description=[G stars] Extended=NO

Proposal 9709 - Observation 1 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 1: TOI-1420 NIRSpec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(TOI-1420 NIRSpec (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	TOI-1420	RA: 21 31 45.9171 (322.9413212d) Dec: +66 20 55.93 (66.34887d) Equinox: J2000			Proper Motion RA: 45.482 mas/yr Proper Motion Dec: 31.874 mas/yr Parallax: 0.0049134" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	306215
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	70	420	1	1	420	26906.242	306215	

Proposal 9709 - Observation 1 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.9746207512218702 to 0.9806106922707393 with period 6.9561063 Days and zero-phase 2459517.43305 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 2 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 2: TOI-1173 b SOSS										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRISS Single-Object Slitless Spectroscopy										
Diagnostics	(TOI-1173 b SOSS (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	TOI-1173b	RA: 13 10 43.7621 (197.6823421d) Dec: +70 46 6.27 (70.76841d) Equinox: J2000			Proper Motion RA: 57.787 mas/yr Proper Motion Dec: -92.39299997716444 mas/yr Parallax: 0.0075664" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems, G stars]										
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	SOSSFAINT	F480M	NISRAPID	15	1	1	0.748	306218	
Template	Subarray					Include Short First Exposure and F277W Exposure?					
	SUBSTRIP256					true					
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	NISRAPID	1	1	1	1	1	11.008			
	2	NISRAPID	4	769	1	1	769	21140.179	306218		
	3	NISRAPID	4	9	1	1	9	247.414			

Proposal 9709 - Observation 2 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.97974 to 0.98564 with period 7.064397357 Days and zero-phase 2458688.715 HJD Aperture PA Range 160 to 180 Degrees (V3 159.43873283 to 179.43873283) Aperture PA Range 225 to 265 Degrees (V3 224.43873283 to 264.43873283) Aperture PA Range 275 to 285 Degrees (V3 274.43873283 to 284.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 3 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 3: TOI-1173 b NIRSpec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(TOI-1173 b NIRSpec (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	TOI-1173b	RA: 13 10 43.7621 (197.6823421d) Dec: +70 46 6.27 (70.76841d) Equinox: J2000			Proper Motion RA: 57.787 mas/yr Proper Motion Dec: -92.39299997716444 mas/yr Parallax: 0.0075664" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems, G stars]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	306218
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	26	867	1	1	867	21132.674	306218	

Proposal 9709 - Observation 3 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.97974 to 0.98564 with period 7.064397357 Days and zero-phase 2458688.71500 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 4 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 4: TOI-2134 c SOSS										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRISS Single-Object Slitless Spectroscopy										
Diagnostics	(TOI-2134 c SOSS (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	TOI-2134_c	RA: 18 07 44.4478 (271.9351992d) Dec: +39 04 26.92 (39.07414d) Equinox: J2000			Proper Motion RA: 54.536 mas/yr Proper Motion Dec: -283.050999905754 mas/yr Parallax: 0.0441087" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	J 6.776 [0.023] C 2003yCat.2246....0C H 6.247 [0.021] C 2003yCat.2246....0C K 6.091 [0.017] C 2003yCat.2246....0C Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	SOSSBRIGHT	F480M	NISRAPID	17	1	1	0.839	306222	
Template	Subarray					Include Short First Exposure and F277W Exposure?					
	SUB204STRIPE_SOSS					true					
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	NISRAPID	1	1	1	1	1	11.461			
	2	NISRAPID	3	1710	1	1	1710	38845.728	306222		
	3	NISRAPID	3	10	1	1	10	227.168			

Proposal 9709 - Observation 4 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.9974382 to 0.9978729 with period 95.852876 Days and zero-phase 2459718.96933 HJD Aperture PA Range 155 to 180 Degrees (V3 154.43873283 to 179.43873283) Aperture PA Range 200 to 220 Degrees (V3 199.43873283 to 219.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 5 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 5: TOI-2134 c NIRSpec											Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Bright Object Time Series											
Diagnostics	(TOI-2134 c NIRSpec (Obs 5)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	TOI-2134_c	RA: 18 07 44.4478 (271.9351992d) Dec: +39 04 26.92 (39.07414d) Equinox: J2000			Proper Motion RA: 54.536 mas/yr Proper Motion Dec: -283.050999905754 mas/yr Parallax: 0.0441087" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	J 6.776 [0.023] C 2003yCat.2246....0C H 6.247 [0.021] C 2003yCat.2246....0C K 6.091 [0.017] C 2003yCat.2246....0C Category=Star Description=[Exoplanet Systems]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	11 TOI_2134c_NIRS pec_TA	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	306225	
Template	Subarray											
	SUB2048											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	G395H/F290LP	NRSRAPID	2	14244	1	1	14244	38835.981	306222		

Proposal 9709 - Observation 5 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.9974382 to 0.9978729 with period 95.852876 Days and zero-phase 2459718.96933 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 6 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 6: WASP-193 SOSS										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRISS Single-Object Slitless Spectroscopy											
Diagnostics	(WASP-193 SOSS (Obs 6)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 6:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	WASP-193	RA: 10 57 23.8522 (164.3493842d) Dec: -29 59 49.67 (-29.99713d) Equinox: J2000			Proper Motion RA: -49.055 mas/yr Proper Motion Dec: 1.506 mas/yr Parallax: 0.0026475" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	H 10.810										
	Category=Star										
	Description=[Exoplanet Systems]										
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	SOSSFAINT	F480M	NISRAPID	9	1	1	0.475	306244	
Template	Subarray					Include Short First Exposure and F277W Exposure?					
	SUBSTRIP256					true					
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	NISRAPID	1	1	1	1	1	11.008			
	2	NISRAPID	12	463	1	1	463	33077.868	306244		
	3	NISRAPID	12	10	1	1	10	714.425			

Proposal 9709 - Observation 6 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.96604 to 0.97271 with period 6.2463345 Days and zero-phase 2457781.66567 HJD Aperture PA Range 310 to 317 Degrees (V3 309.43873283 to 316.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 7 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 7: WASP-127b NIRSpec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(WASP-127b NIRSpec (Obs 7)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	WASP-127	RA: 10 42 14.0837 (160.5586821d) Dec: -03 50 6.26 (-3.83507d) Equinox: J2000			Proper Motion RA: 19.133 mas/yr Proper Motion Dec: 17.058 mas/yr Parallax: 0.0062241" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	12 WASP-127_NIRSpec_TA	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	306249
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	14	2754	1	1	2754	37318.022	306251	

Proposal 9709 - Observation 7 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.9433244572718531 to 0.9532971755924664 with period 4.178062 Days and zero-phase 2457248.741 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 8 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 8: KELT-11 SOSS									Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRISS Single-Object Slitless Spectroscopy									
Diagnostics	(KELT-11 SOSS (Obs 8)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.									
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(6)	KELT-11	RA: 10 46 49.7402 (161.7072508d) Dec: -09 23 56.50 (-9.39903d) Equinox: J2000		Proper Motion RA: -79.16 mas/yr Proper Motion Dec: -78.50699996652111 mas/yr Parallax: 0.010004899999999999" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Star Description=[Exoplanet Systems]									
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	SOSSBRIGHT	F480M	NISRAPID	9	1	1	0.475	306256
Template	Subarray					Include Short First Exposure and F277W Exposure?				
	SUB204STRIPE_SOSS					true				
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	NISRAPID	1	1	1	1	1	11.461		
	2	NISRAPID	3	2278	1	1	2278	51748.87	306256	
	3	NISRAPID	3	10	1	1	10	227.168		

Proposal 9709 - Observation 8 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.9322944756469348 to 0.9410919552971063 with period 4.73620495 Days and zero-phase 2458260.168608 HJD Aperture PA Range 105 to 120 Degrees (V3 104.43873283 to 119.43873283) Aperture PA Range 280 to 320 Degrees (V3 279.43873283 to 319.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 16 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 16: KELT-11 NIRSpec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(KELT-11 NIRSpec (Obs 16)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(KELT-11 NIRSpec (Obs 16)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 51.025 Arcsec (larger than the recommended limit of 50.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.										
	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	KELT-11	RA: 10 46 49.7402 (161.7072508d) Dec: -09 23 56.50 (-9.39903d) Equinox: J2000			Proper Motion RA: -79.16 mas/yr Proper Motion Dec: -78.50699996652111 mas/yr Parallax: 0.010004899999999999" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	13 KELT_11_NIRSpec_TA	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	306255
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	2	18975	1	1	18975	51734.958	306256	

Proposal 9709 - Observation 16 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.9322944756469348 to 0.9410919552971063 with period 4.73620495 Days and zero-phase 2458260.168608 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 9 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Observation	Proposal 9709, Observation 9: WASP-131 SOSS									Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRISS Single-Object Slitless Spectroscopy									
Diagnostics	(WASP-131 SOSS (Obs 9)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.									
	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 9:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(7)	WASP-131	RA: 14 00 46.4448 (210.1935200d) Dec: -30 35 0.85 (-30.58357d) Equinox: J2000		Proper Motion RA: 10.24 mas/yr Proper Motion Dec: -9.20499992389523 mas/yr Parallax: 0.0051765" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Star Description=[Exoplanet Systems]									
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	SOSSFAINT	F480M	NISRAPID	11	1	1	0.566	306412
Template	Subarray					Include Short First Exposure and F277W Exposure?				
	SUB680STRIPE_SOSS					true				
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	NISRAPID	1	1	1	1	1	11.092		
	2	NISRAPID	4	1086	1	1	1086	30015.563	306412	
	3	NISRAPID	4	10	1	1	10	276.386		

Proposal 9709 - Observation 9 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the m...

Special Requirements	Phase 0.9634471906817483 to 0.9712763092091224 with period 5.3220125 Days and zero-phase 2458282.2588 HJD Aperture PA Range 87 to 89 Degrees (V3 86.43873283 to 88.43873283) Aperture PA Range 297 to 305 Degrees (V3 296.43873283 to 304.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 10 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 10: WASP-131 NIRspec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRSpec Bright Object Time Series											
Diagnostics	(WASP-131 NIRspec (Obs 10)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	WASP-131	RA: 14 00 46.4448 (210.1935200d) Dec: -30 35 0.85 (-30.58357d) Equinox: J2000			Proper Motion RA: 10.24 mas/yr Proper Motion Dec: -9.20499992389523 mas/yr Parallax: 0.0051765" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	14 WASP_131_NIR Spec_TA	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	306419
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	10	3018	1	1	3018	30006.405	306412	

Proposal 9709 - Observation 10 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.9634471906817483 to 0.9712763092091224 with period 5.3220125 Days and zero-phase 2458282.2588 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 11 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 11: WASP-20 SOSS									Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRISS Single-Object Slitless Spectroscopy									
Diagnostics	(WASP-20 SOSS (Obs 11)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.									
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(8)	WASP-20	RA: 00 20 38.5350 (5.1605625d) Dec: -23 56 8.60 (-23.93572d) Equinox: J2000		Proper Motion RA: 1.352 mas/yr Proper Motion Dec: -14.739000062036212 mas/yr Parallax: 0.0034699" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Star Description=[Exoplanet Systems]									
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	SOSSFAINT	F480M	NISRAPID	7	1	1	0.384	306434
Template	Subarray					Include Short First Exposure and F277W Exposure?				
	SUBSTRIP256					true				
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	NISRAPID	1	1	1	1	1	11.008		
	2	NISRAPID	4	902	1	1	902	24796.413	306434	
	3	NISRAPID	4	10	1	1	10	274.905		

Proposal 9709 - Observation 11 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.96747 to 0.97597 with period 4.89964477 Days and zero-phase 2458087.08446 HJD Aperture PA Range 30 to 70 Degrees (V3 29.43873283 to 69.43873283) Aperture PA Range 235 to 280 Degrees (V3 234.43873283 to 279.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 12 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 12: WASP-20 NIRSpec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(WASP-20 NIRSpec (Obs 12)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(8)	WASP-20	RA: 00 20 38.5350 (5.1605625d) Dec: -23 56 8.60 (-23.93572d) Equinox: J2000			Proper Motion RA: 1.352 mas/yr Proper Motion Dec: -14.739000062036212 mas/yr Parallax: 0.0034699" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	306434
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	30	885	1	1	885	24764.495	306434	

Proposal 9709 - Observation 12 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.96747 to 0.97597 with period 4.89964477 Days and zero-phase 2458087.08446 HJD Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 13 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 13: HD 209458b SOSS									Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRISS Single-Object Slitless Spectroscopy									
Diagnostics	(HD 209458b SOSS (Obs 13)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.									
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(9)	HD-209458b	RA: 22 03 10.7727 (330.7948863d) Dec: +18 53 3.55 (18.88432d) Equinox: J2000		Proper Motion RA: 29.76599999999995 mas/yr Proper Motion Dec: -17.976000003727677 mas/yr Parallax: 0.0207694" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Star Description=[Exoplanet Systems]									
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	SOSSBRIGHT	F480M	NISRAPID	7	1	1	0.384	306437
Template	Subarray					Include Short First Exposure and F277W Exposure?				
	SUB60STRIPE_SOSS					true				
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	NISRAPID	1	1	1	1	1	12.724		
	2	NISRAPID	5	676	1	1	676	24863.172	306437	
	3	NISRAPID	5	10	1	1	10	367.798		

Proposal 9709 - Observation 13 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.953268335278042 to 0.9650895049634902 with period 3.52474859 Days and zero-phase 2451659.93742 HJD Aperture PA Range 90 to 110 Degrees (V3 89.43873283 to 109.43873283) Aperture PA Range 200 to 220 Degrees (V3 199.43873283 to 219.43873283) Time Series Observation No Parallel Attachments
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Proposal 9709 - Observation 15 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 15: KELT-4 NIRSpec										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(KELT-4 NIRSpec (Obs 15)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(KELT-4 NIRSpec (Obs 15)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 48.774 Arcsec (larger than the recommended limit of 40.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.										
	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	KELT-4	RA: 10 28 15.0111 (157.0625463d) Dec: +25 34 23.47 (25.57319d) Equinox: J2000			Proper Motion RA: 14.118341354376284 mas/yr Proper Motion Dec: -12.11922514602998 mas/yr Parallax: 0.00458" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Parallax, PM from Gaia DR3 Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	15 KELT_4_NIRSpec_TA	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	306475
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSRAPID	20	1410	1	1	1410	26737.097	306472	

Proposal 9709 - Observation 15 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.9412424146364888 to 0.955179680611814 with period 2.98958682 Days and zero-phase 2456617.81302 HJD Aperture PA Range 105 to 135 Degrees (V3 326.14704895 to 356.14704895) Aperture PA Range 270 to 300 Degrees (V3 131.14704895 to 161.14704895) Time Series Observation No Parallel Attachments Sequence Observations 15, 17, Non-interruptible
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Proposal 9709 - Observation 17 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 17: NIRISS-Offset_test_15										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRISS Single-Object Slitless Spectroscopy										
Diagnostics	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	KELT-4	RA: 10 28 15.0111 (157.0625463d) Dec: +25 34 23.47 (25.57319d) Equinox: J2000			Proper Motion RA: 14.118341354376284 mas/yr Proper Motion Dec: -12.11922514602998 mas/yr Parallax: 0.00458" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Parallax, PM from Gaia DR3 Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	SOSSFAINT	F480M	NISRAPID	11	1	1	0.566	172682	
Template	Subarray					Include Short First Exposure and F277W Exposure?					
	SUB680STRIPE_SOSS					false					
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	NISRAPID	5	30	1	1	30	994.622	172682		

Proposal 9709 - Observation 17 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Aperture PA Range 108 to 136 Degrees (V3 107.43873283 to 135.43873283) Offset 14.734360827453132 arcsec, 0.0 arcsec Time Series Observation No Parallel Attachments 14 After 17 Sequence Observations 15, 17, Non-interruptible
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Proposal 9709 - Observation 14 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Observation	Proposal 9709, Observation 14: KELT-4 SOSS										Thu Jun 18 17:00:20 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRISS Single-Object Slitless Spectroscopy											
Diagnostics	(KELT-4 SOSS (Obs 14)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	KELT-4	RA: 10 28 15.0111 (157.0625463d) Dec: +25 34 23.47 (25.57319d) Equinox: J2000			Proper Motion RA: 14.118341354376284 mas/yr Proper Motion Dec: -12.11922514602998 mas/yr Parallax: 0.00458" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Parallax, PM from Gaia DR3 Category=Star Description=[Exoplanet Systems]										
Acquisition	#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	SOSSFAINT	F480M	NISRAPID	11	1	1	0.566	306472	
Template	Subarray					Include Short First Exposure and F277W Exposure?					
	SUB680STRIPE_SOSS					true					
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	NISRAPID	1	1	1	1	1	11.092			
	2	NISRAPID	5	807	1	1	807	26755.343	306472		
	3	NISRAPID	5	10	1	1	10	331.541			

Proposal 9709 - Observation 14 - Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the ...

Special Requirements	Phase 0.9412424146364888 to 0.955179680611814 with period 2.98958682 Days and zero-phase 2456617.81302 HJD Aperture PA Range 105 to 135 Degrees (V3 104.43873283 to 134.43873283) Aperture PA Range 270 to 300 Degrees (V3 269.43873283 to 299.43873283) Time Series Observation No Parallel Attachments 14 After 17
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